

Chapter 6, Problem 1PP

Problem

What is the voltage across a $4.5\text{-}\mu\text{F}$ capacitor if the charge on one plate is 0.12 mC ? How much energy is stored?

Step-by-step solution

Step 1 of 2

Write the formula for the charge on the plate.

$$q = Cv$$

The voltage across the plates is,

$$v = \frac{q}{C}$$

Substitute 4.5×10^{-6} for C and 0.12×10^{-3} for q .

$$\begin{aligned} v &= \frac{0.12 \times 10^{-3}}{4.5 \times 10^{-6}} \\ &= 26.67 \text{ V} \end{aligned}$$

Thus, the voltage across the capacitor is $\boxed{26.67 \text{ V}}$.

Step 2 of 2

Write the formula for energy stored in the capacitor.

$$w = \frac{1}{2} Cv^2$$

Substitute 4.5×10^{-6} for C and 26.67 for v .

$$\begin{aligned} w &= \left(\frac{1}{2}\right)(4.5 \times 10^{-6})(26.67)^2 \\ &= 1.6 \times 10^{-3} \text{ J} \\ &= 1.6 \text{ mJ} \end{aligned}$$

Thus, the energy stored in the capacitor is $\boxed{1.6 \text{ mJ}}$.